



Summer Term 2026

Investment and Financial Management
Exercise on Mathematical Basics

Session 07: Bonds I

Center for Digital Transformation

Technische Universität München (TUM), Campus Heilbronn

Recap

Bonds are long-term contracts with issuers (borrowers) committing themselves to repay debt at a specified date and, if applicable, to pay coupon payments at predetermined points in time to the bond owner

- Since a bond is a debt instrument, usually its cash flows can be easily predicted.
 - Exception: variable coupon
 - Insolvency risk (credit risk)
- The discount or interest rate applied in the pricing formula equals the return of an alternative investment with a similar level of risk.

Pricing Principle

As real investments, financial investments can also be priced using the ***Discounted Cash Flow-Principle***.

4.1 Introduction

4.2 Bond pricing

4.2.1 Pricing of zerobonds

4.2.2 Pricing of coupon bonds

4.3 Features of bonds

4.3.1 Bond pricing with sloping yields curves

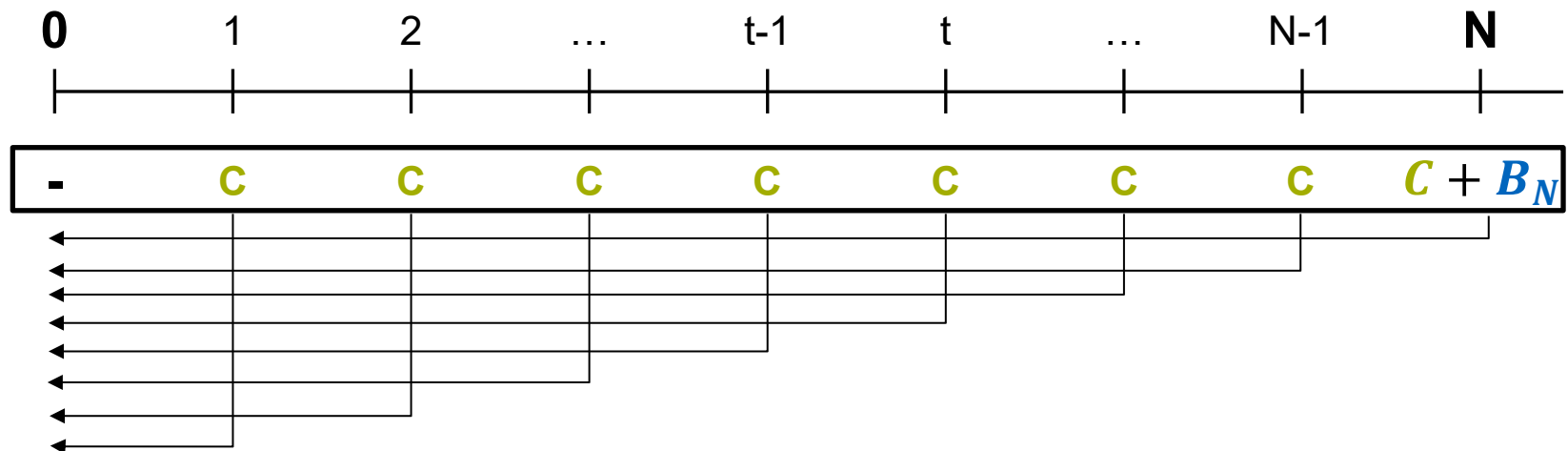
4.3.2 Risk of bonds

4.3.3 Concept of duration

4.4 Exercises

A coupon bond pays regular interest payments at specified points in time. The principal amount is paid at maturity of the bond.

To determine the price of a coupon bond all payments – **coupon payments** and **redemption payment** – are discounted with the risk-adjusted interest rate. This calculation method is equal to the valuation of an annuity-immediate with C and $C + B_N$ as annuity payments.



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This calculation method is equal to the valuation of an annuity-immediate with C and $C + B_N$ as annuity payments.

$$(F): B_0 = \sum_{k=1}^N \frac{C}{(1+r)^k} + \frac{B_N}{(1+r)^N}$$

(A.3) What is the price of a coupon bond with a coupon of 10% and a face value of 100 EUR if the interest rate is assumed to be 10%, assuming the maturity is 3 years?

$$(1): B_0 = \sum_{k=1}^N \frac{C}{(1+r)^k} + \frac{B_N}{(1+r)^N}$$

$$(2): B_0 = \frac{10}{1.1^1} + \frac{10}{1.1^2} + \frac{110}{1.1^3} = 100 \text{ EUR}$$

A coupon bond pays regular interest payments at specified points in time. The principal amount is paid at maturity of the bond.

$$(1): B_0 = \sum_{k=1}^N \frac{C}{(1+r)^k} + \frac{B_N}{(1+r)^N}$$

$$(2): B_0 = \left[C * \frac{(1+r)^N - 1}{r} + B_N \right] * \frac{1}{(1+r)^N}$$

By simplification and the use of the future value factor we can derive...

Categorization of bonds: Price of the bond as % of nominal value

- $B_0 = 100$: *Bond is traded at par*
- $B_0 < 100$: *Bond is traded below par (at discount)*
- $B_0 > 100$: *Bond is traded above par (at premium)*

(A.3) The bond of BMW AG is a coupon bond with an annual coupon of 6% which will be repaid after 10 years at par (100 EUR). What will be its price if we assume an effective annual interest rate of 6.5%?

$$(1): B_0 = \left[C * \frac{(1 + r)^N - 1}{r} + B_N \right] * \frac{1}{(1 + r)^N}$$

$$(2): B_0 = \left[6 * \frac{(1 + 0.065)^{10} - 1}{0.065} + 100 \right] * \frac{1}{(1 + 0.065)^{10}} = 96.41 \text{ EUR (discount)}$$

Variation of the interest rate

- $r = 0.055 \Rightarrow B_0 = 103.77 \text{ EUR (premium)}$
- $r = 0.06 \Rightarrow B_0 = 100 \text{ EUR (par)}$

(A.4) A bond with a coupon of 6% has a time to maturity of 20 years and is repaid with its face value at the end of this period. What is the price of the bond at an effective annual interest rate of 6.2%?

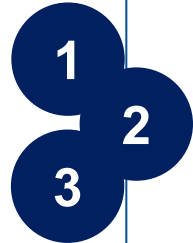
$$(1): B_0 = \left[C * \frac{(1 + r)^N - 1}{r} + B_N \right] * \frac{1}{(1 + r)^N}$$

$$(2): B_0 = \left[6 * \frac{(1 + 0.062)^{20} - 1}{0.062} + 100 \right] * \frac{1}{(1 + 0.062)^{20}} = 97.74 \text{ EUR}$$

★ **(A.5)** The effective annual interest rate equals 4%. The Wailua AG issues a coupon bond with a maturity of 10 years. After ten years the face value of 100 EUR will be repaid. The coupon is 6%.

(a) What will be the issue price?

(b) After 5 years the market interest rates goes up to 5%. What is the market value of the bond now?



$$B_0 = 104.33 \text{ EUR};$$

$$B_5 = 100.00 \text{ EUR}$$

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$$B_5 = 104.33 \text{ EUR}$$

$$(1): B_0 = \left[C * \frac{(1 + r)^N - 1}{r} + B_N \right] * \frac{1}{(1 + r)^N}$$

$$(2): B_0 = \left[6 * \frac{(1 + 0.04)^{10} - 1}{0.04} + 100 \right] * \frac{1}{(1 + 0.04)^{10}} = 116.22 \text{ EUR}$$

$$(3): B_5 = \left[6 * \frac{(1 + 0.05)^5 - 1}{0.05} + 100 \right] * \frac{1}{(1 + 0.05)^5} = 104.33 \text{ EUR}$$

(A.5) The effective annual interest rate equals 4%. The Wailua AG issues a coupon bond with a maturity of 10 years. After ten years the face value of 100 EUR will be repaid. The coupon is 6%.

- (a) What will be the issue price?
- (b) After 5 years the market interest rates goes up to 5%. What is the market value of the bond now?
- (c) What is the average annual return for an investor who has initially bought the bond and sells it after five years? (assume reinvestment at market interest rate)**

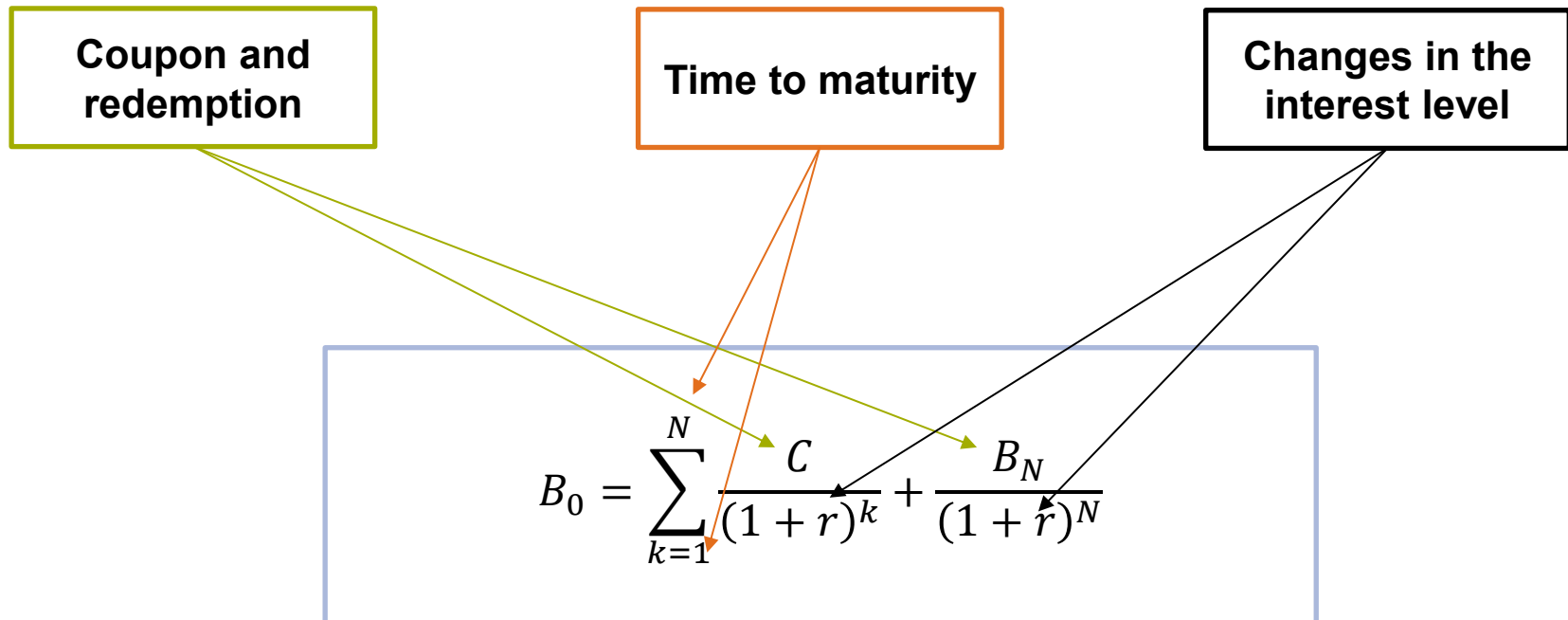
$$(1): B_5 = 104.33 \text{ EUR}, B_0 = 116.22 \text{ EUR}$$

$$(2): FV_5^C = 6 * 1.04^4 + 6 * 1.04^3 + 6 * 1.04^2 + 6 * 1.04^1 + 6 = 32.50 \text{ EUR}$$

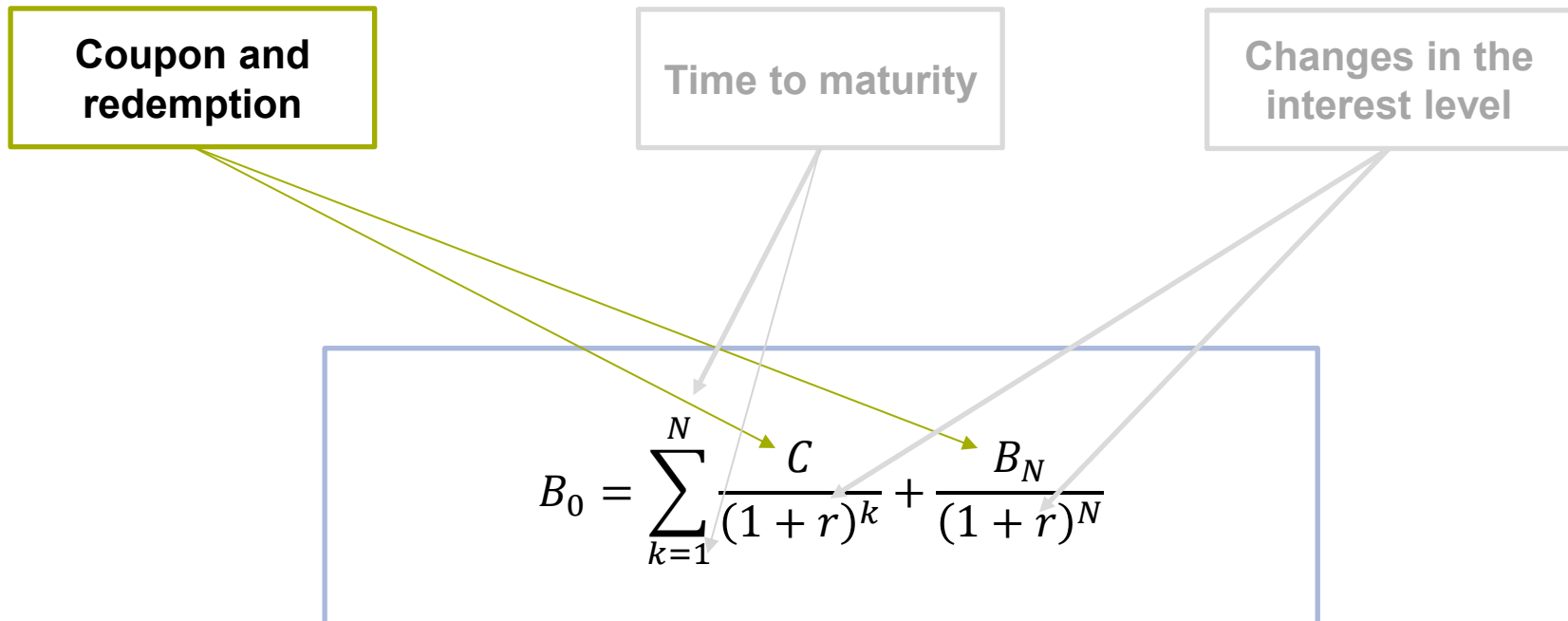
$$(3): q = \sqrt[5]{\frac{104.33 + 32.50}{116.22}} = 1.0332 \Rightarrow r = 0.0332 = 3.32\% < 4\%$$

Due to the change in the interest level, the effective interest on the bond is lower than the market interest rate.

Factors influencing the pricing of a coupon bond...



Factors influencing the pricing of a coupon bond...



The higher the coupon and the redemption payment, the higher the market price of the underlying coupon bond

Concept of accrued interest:

- If the bond ownership changes between two interest payments, the buyer needs to pay a compensation to the seller due to the proportionate interest claims the seller has gained.
- The buying price will then equal the sum of market price and accrued interests.

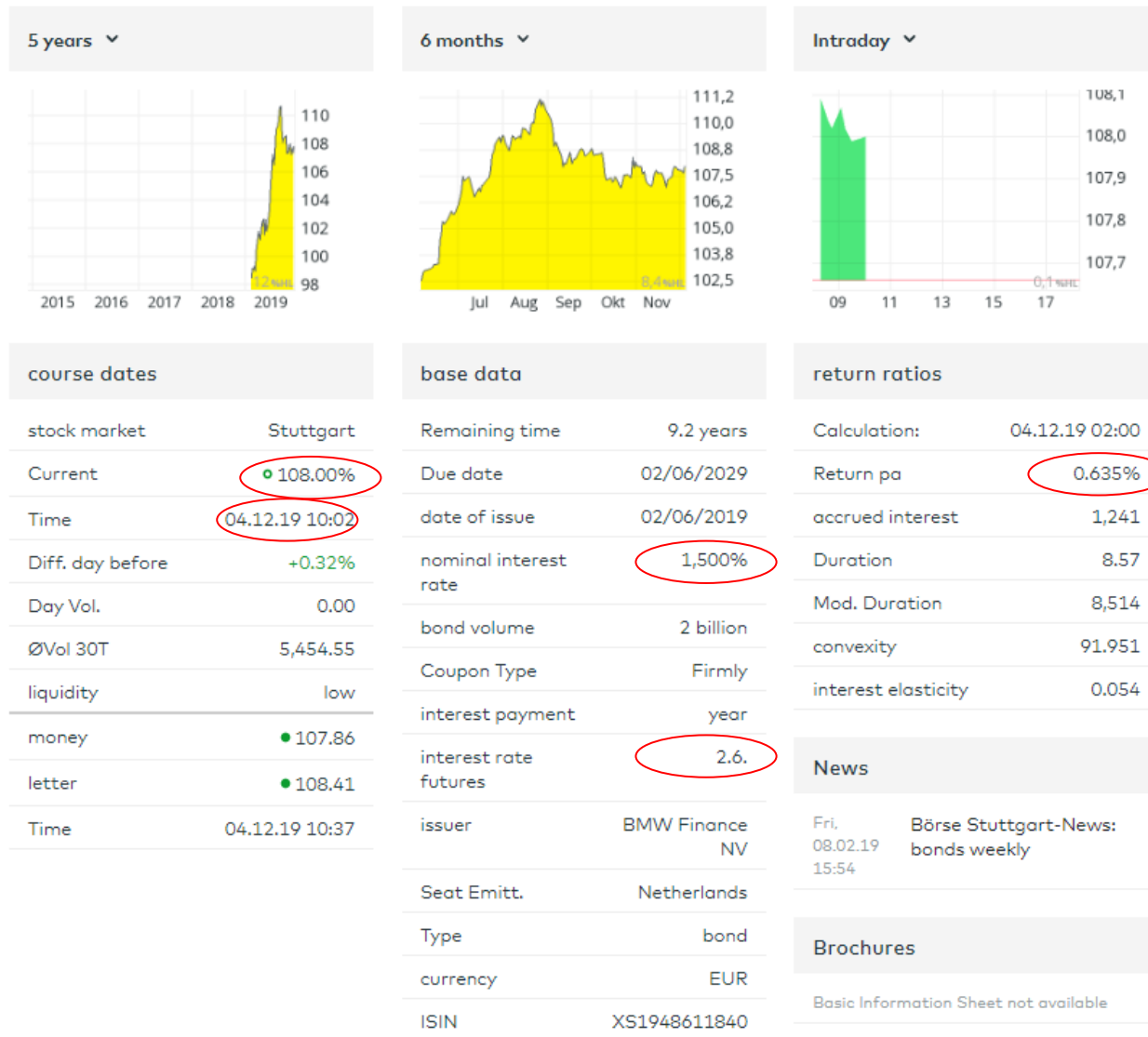
Settlement of fixed-income securities at financial markets

- Interest claims between last interest payment date and selling date are added to the market price of the security
- $I_0 = (t_1 - t_0) * \frac{1}{360} * C$
- I_0 – Interest claims, with t_1 – selling date, and t_0 – last interest payment

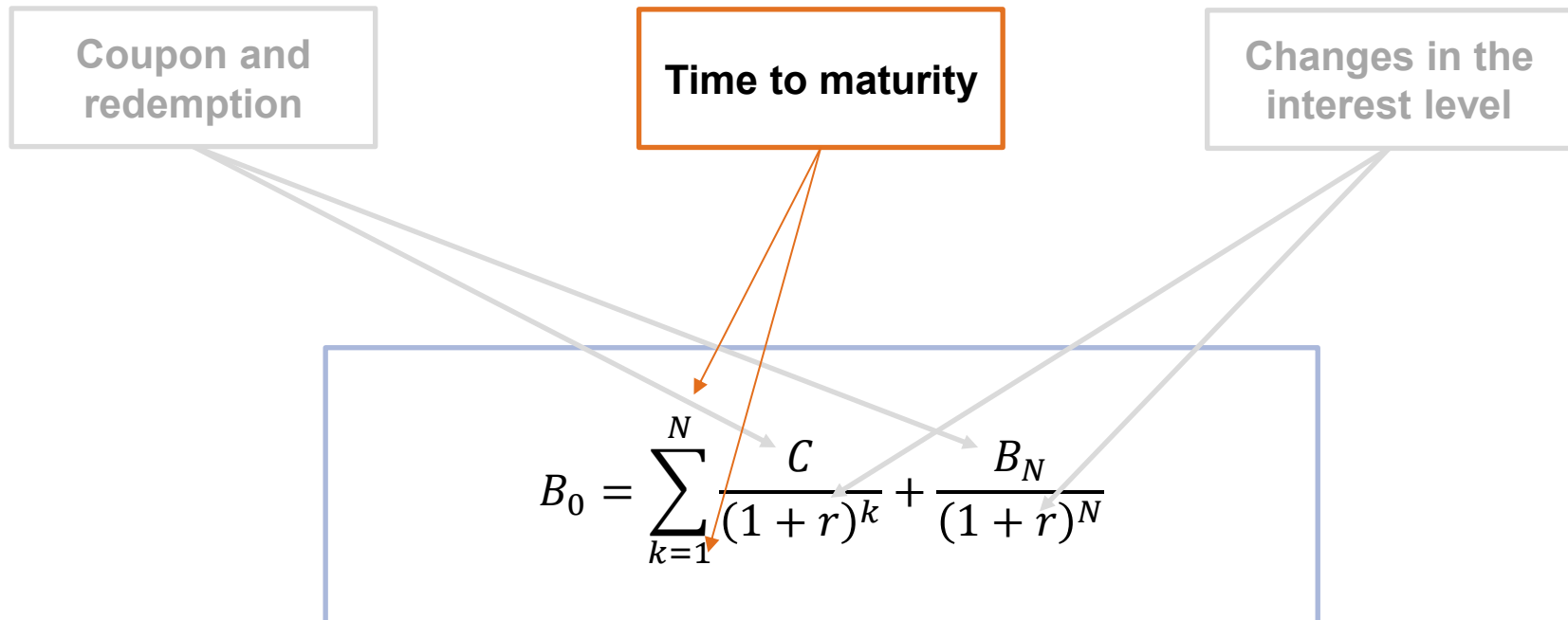
The higher the coupon and the redemption payment, the higher the market price of the underlying coupon bond

- **Important:** Due to the accrued interests, the selling price does not automatically equal the market price.
- At exchanges only the market price is listed: “fair” price between buyer and seller without the consideration of accrued interests (“ex coupon price”)
- **Goal:** Avoiding price variation just due to changes of accrued interests
- The final selling price is the sum of the market price, accrued interests, transaction costs, merchant fees, and brokerage fees.
- Due to simplicity we only focus on market price and accrued interests

Factors influencing the market price of a coupon bond



Factors influencing the pricing of a coupon bond...



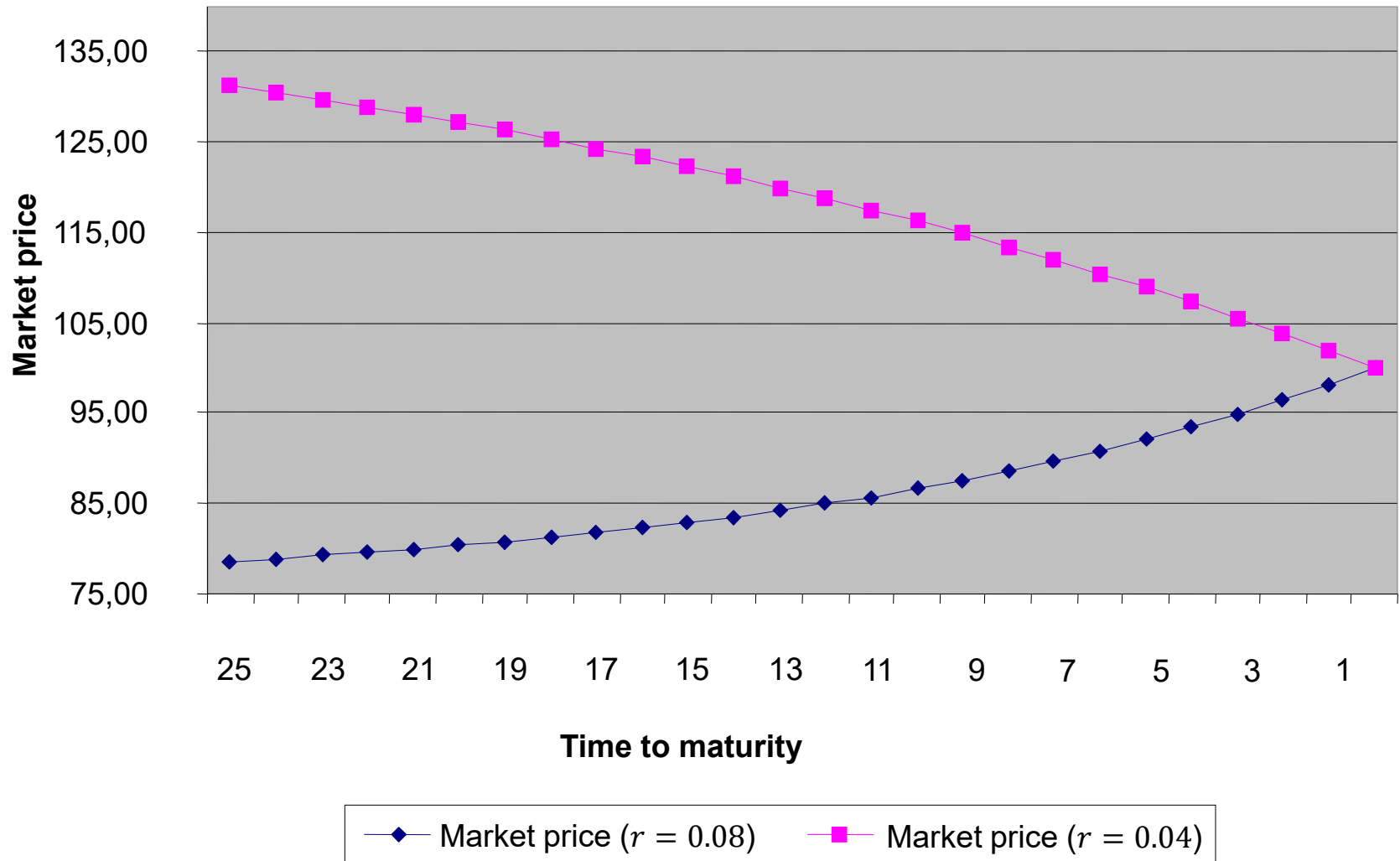
The market price of a bond is dependent on its time to maturity!

Let's take the example (A.3):

The BMW coupon bond with a coupon of 6% is issued. The risk-adjusted interest rate should equal (a) 4% or (b) 8%. What does happen to the market value if the time to maturity varies?

Time to maturity	Market Prices		
	$r = 0.08$	$r = 0.06$	$r = 0.04$
5	92.0146	100	108.9036
10	86.5798	100	116.2218
15	82.8810	100	122.2368
20	80.3637	100	127.1807
25	78.6504	100	131.2442

Factors influencing the market price of a coupon bond



The market price of a bond is dependent on its time to maturity!

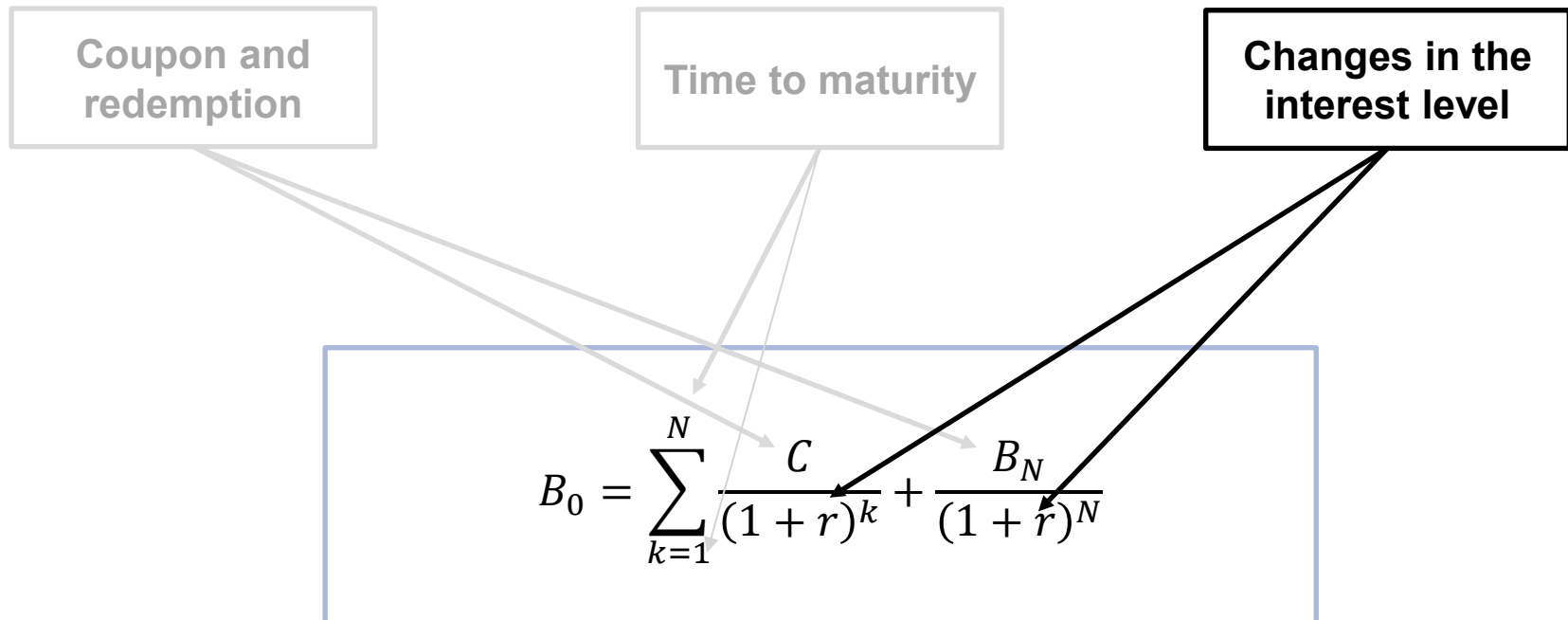
Interpretation of the graph:

- At maturity the price of a bond equals its nominal (face) value.
- The market price of a discount bond (below par) increases to the face value (convex behavior of the present value function).
- The market price of a premium bond (above par) decreases to the face value (concave behavior of the present value function).

The concept of Yield to Maturity (YTM)

- Yield to Maturity defines the return that will be earned by the bond owner when the bond is held until maturity under the condition that the risk-adjusted market interest rate remains unchanged (re-investment hypothesis). The Yield to Maturity is denoted per annum (p.a.).

Factors influencing the pricing of a coupon bond...



Bond prices are sensitive to changes in the interest rate!

- To be precise: Those bond owners are exposed to interest rate risk who want to sell their bonds before maturity.
- The lower the market interest rate, the higher the market price of a bond because the present value of the single payments of the bond increases.

- Let's look at the example:

- The BMW coupon bond has a coupon of 6%.

- Case I: $r = 0.055 \Rightarrow B_0 = 103.77 \text{ EUR}$

- Case II: $r = 0.060 \Rightarrow B_0 = 100.00 \text{ EUR}$

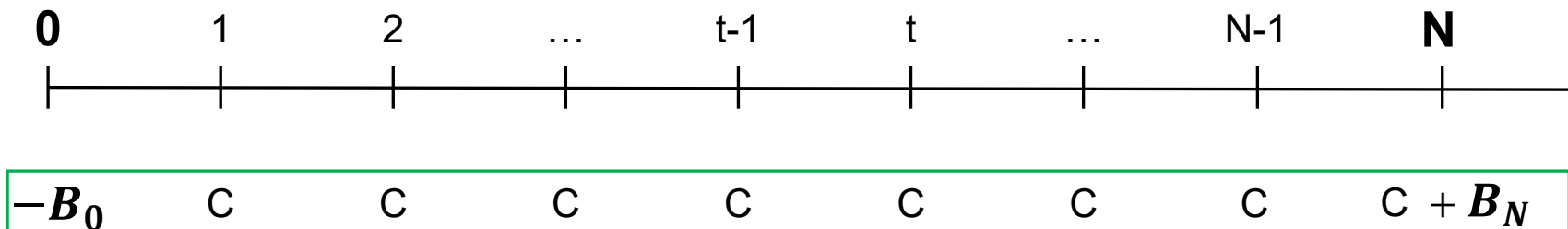
- Case III: $r = 0.065 \Rightarrow B_0 = 96.41 \text{ EUR}$

- Conclusion:

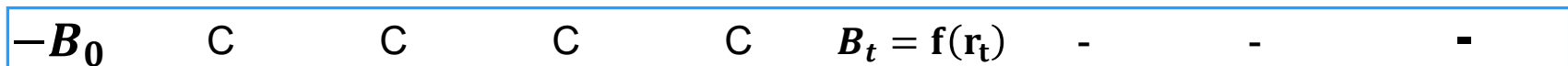
- If the market interest rate falls below the coupon rate, the price of the bond increases above par.
 - If the market interest increases above the coupon rate, the price of the bond decreases below par.

Bond prices are sensitive to changes in the interest rate!

- To be precise: Those bond owners are exposed to interest rate risk who want to sell their bonds before maturity.
- Remember: To calculate the return of an investment, we look at the cash flows of the underlying investment.

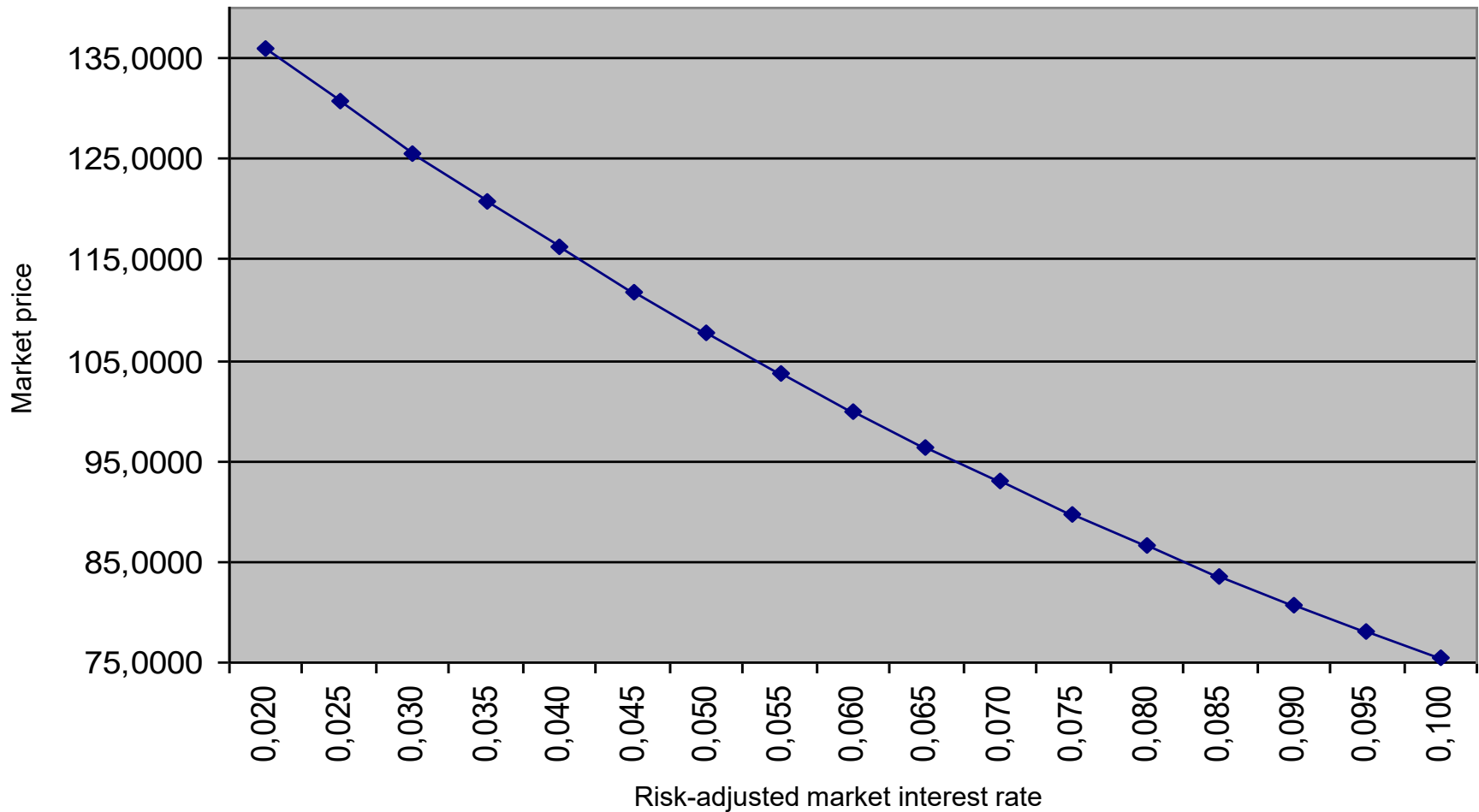


If the bond is held until maturity, no cash flow depends on the interest rate.



If the bond is sold before maturity, the selling price at t strongly depends on the current interest rate at t.

Bond prices are sensitive to changes in the interest rate!



(A.7) The corporate bond of Allianz AG with a nominal value of 100 EUR has a time to maturity of 4 years. The yearly coupon payments pay off 8% of the face value. The redemption amount will equal 101 EUR (1 EUR as premium). The market interest rate for this risk class is 8.5% p.a.

(a) What is the market value of the bond?

(b) What will be the future wealth of an investor who will buy the bond today, will hold it until maturity, and will reinvest all coupon payments at the market interest rate?

$$(a1): B_0 = \left[C * \frac{(1 + r)^N - 1}{r} + B_N \right] * \frac{1}{(1 + r)^N}$$

$$(a2): B_0 = \left[8 * \frac{(1 + 0.085)^4 - 1}{0.085} + 101 \right] * \frac{1}{(1 + 0.085)^4} = 99.08 \text{ EUR}$$

(A.7) The corporate bond of Allianz AG with a nominal value of 100 EUR has a time to maturity of 4 years. The yearly coupon payments pay off 8% of the face value. The redemption amount will equal 101 EUR (1 EUR as premium). The market interest rate for this risk class is 8.5% p.a.

(a) What is the market value of the bond?

(b) What will be the future wealth of an investor who will buy the bond today, will hold it until maturity, and will reinvest all coupon payments at the market interest rate?

$$(b1): FV = B_N + \sum_{k=1}^N C * (1 + r)^{N-k}$$

$$(b2): FV = 101 \text{ EUR} + 8 * (1.085)^3 + 8 * (1.085)^2 + 8 * (1.085)^1 + 8 = 137.32 \text{ EUR}$$

$$(b3): r^* = \sqrt[4]{\frac{137.32 \text{ EUR}}{99.08 \text{ EUR}}} - 1 = 8.50\% = \text{Yield to Maturity}$$

★ (A.8) Fiat AG issues a 10-year bond with a coupon of 5%. The interest rate for 10-years government bonds equals 5.75%, the company has an A rating. Such securities are traded at financial markets with a risk premium of 50 basis points. What will be the issue price of the bond? What would be the price if the bond would not pay any coupon payments (zerobond)?

1

2

3

$$B_0 = 75.03 \text{ EUR}; \\ B_0^{ZB} = 100.00 \text{ EUR}$$

$$B_0 = 104.33 \text{ EUR}; \\ B_0^{ZB} = 111.23 \text{ EUR}$$

$$B_0 = 100.00 \text{ EUR}; \\ B_0^{ZB} = 99.12 \text{ EUR}$$

$$B_0 = 90.91 \text{ EUR}; \\ B_0^{ZB} = 54.54 \text{ EUR}$$

$$(1): B_0 = \left[C * \frac{(1 + r)^N - 1}{r} + B_N \right] * \frac{1}{(1 + r)^N}$$

$$(2): B_0 = \left[5 * \frac{(1 + 0.0625)^{10} - 1}{0.0625} + 100 \right] * \frac{1}{(1 + 0.0625)^{10}} = 90.91 \text{ EUR}$$

$$(3): B_0 = B_N * q^{-N} = 100 * 1.0625^{-10} = 54.54 \text{ EUR}$$